

Pricing American Put Options with Least-Squares Monte Carlo: A Study of Longstaff–Schwartz and Tsitsiklis–Van Roy

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1 Introduction

American option pricing is fundamentally an optimal stopping problem. Unlike a European claim, an American contract can be exercised prior to maturity, so valuation requires not only a price but also a rule describing when exercise is optimal. This feature makes American-style derivatives substantially more difficult to value than European derivatives. Even in the simple case of a vanilla American put under geometric Brownian motion, valuation cannot be reduced to a single terminal expectation, because the holder must repeatedly compare the payoff from immediate exercise with the value of waiting.

This report studies that problem through two influential simulation-based papers. Longstaff and Schwartz (2001) develop the least-squares Monte Carlo approach, whose key idea is to estimate the conditional expected payoff from continuation by least-squares regression on simulated paths. Their method is motivated by the observation that, at each exercise date, the optimal decision depends on comparing immediate exercise value with the conditional expectation of future discounted cash flows from continuation. By estimating that conditional expectation from cross-sectional path information, the method converts a difficult optimal stopping problem into a sequence of regression problems.

Tsitsiklis and Van Roy (2001) study the same class of American-style pricing problems from the viewpoint of approximate dynamic programming. Their framework starts from the Bellman recursion and emphasizes recursive approximation of the value function over a parameterized function class. In this formulation, the central issue is not only how to estimate continuation values, but also how approximation error propagates backward through time. The paper shows that naive approximation schemes can generate errors that grow rapidly with the horizon, while trajectory-based sampling under the risk-neutral distribution yields a much more stable approximation framework.

The two papers therefore address the same computational challenge but from different angles. Longstaff–Schwartz is more directly tied to continuation-value regression and practical pathwise stopping-rule construction, whereas Tsitsiklis–Van Roy is more explicitly framed in terms of value iteration, projection, and approximation error control. In the present project, Longstaff–Schwartz is treated as the main practical pricing method, and Tsitsiklis–Van Roy is implemented as a matched comparator under the same computational testbeds. This shared design allows differences in the results to be interpreted as methodological differences rather than differences in case construction.

The empirical study is organized around a shared synthetic benchmark rather than historical equity calibration. The main benchmark is a vanilla American put under risk-neutral geometric Brownian motion, priced against an implicit finite-difference American benchmark, a Black–Scholes European lower reference, and a CRR tree check. This shared benchmark is then extended in two directions: an Asian-style extension with multiple state variables and a jump-style extension showing how both regression methods behave when the state dynamics change.

The purpose of the project is to study the two papers with implementation evidence, testing, and critical assessment. It is not a claim of exact numerical reproduction of every published table. The emphasis is on understanding the pricing logic, implementing it in a coherent way, and comparing the two approaches under a common design.

2 Problem setup: American option pricing as an optimal stopping problem

Let $(\Omega, \mathcal{F}, \mathbb{Q})$ be a risk-neutral probability space equipped with a filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$ generated by the relevant state variables of the pricing model. For an American put with strike K and maturity T written on an underlying process $(S_t)_{0 \leq t \leq T}$, the holder may choose an admissible stopping time τ taking values in $[0, T]$. Under no-arbitrage pricing, the time-zero value is

$$V_0 = \sup_{\tau \in \mathcal{T}} \mathbb{E}^{\mathbb{Q}} [e^{-r\tau} (K - S_{\tau})^+],$$

where r is the continuously compounded risk-free rate and $\mathcal{T}$ is the set of stopping times with respect to the filtration. This formulation makes clear that American option valuation is not only a pricing problem but also a decision problem: one must determine the exercise rule that maximizes expected discounted payoff. This is the standard optimal stopping representation underlying both Longstaff–Schwartz and Tsitsiklis–Van Roy.

In practice, numerical valuation is usually carried out in discrete time. If the allowable exercise dates are

$$0 = t_0 < t_1 < \dots < t_N = T,$$

then the American option is approximated by a Bermudan option that may only be exercised on this grid. This restriction slightly lowers the value relative to the continuous-time contract, but under standard regularity conditions the approximation improves as the exercise mesh becomes finer. Tsitsiklis and Van Roy explicitly adopt this discrete-time setup in order to formulate the pricing problem through dynamic programming, and Longstaff–Schwartz likewise implement their simulation method on a finite exercise grid.

Under this discrete-time formulation, the Bellman recursion is

$$V_i(S_{t_i}) = \max \{h(S_{t_i}), C_i(S_{t_i})\},$$

where

$$h(S_{t_i}) = (K - S_{t_i})^+$$

is the intrinsic value and

$$C_i(S_{t_i}) = \mathbb{E}^{\mathbb{Q}} [e^{-r\Delta t} V_{i+1}(S_{t_{i+1}}) \mid S_{t_i}]$$

is the continuation value. The continuation term is the conditional expected discounted value of waiting one more step and then acting optimally thereafter. American pricing therefore requires repeated comparison of immediate exercise and continuation value at every exercise date.

This recursion also explains why American option pricing is much harder than European pricing. For a European option, valuation reduces to one terminal expectation under the risk-neutral measure. For an American option, the continuation value itself is an unknown conditional expectation embedded inside a backward recursion with a max operator. In low-dimensional settings, this recursion can be handled directly by trees or finite-difference methods. In higher-dimensional settings, however, explicit state-space discretization becomes computationally prohibitive. The contribution of both Longstaff–Schwartz and Tsitsiklis–Van Roy is to replace dense state-space discretization with regression-based approximations built from simulated paths.

For the purposes of this report, this setup also clarifies the methodological difference between the two papers. Longstaff–Schwartz approximates the continuation value by regressing discounted realized future cash flows on basis functions of the current state. Tsitsiklis–Van Roy instead frame the problem as approximate value iteration and recursively approximate the value function, or equivalently the continuation component, through projection onto a finite-dimensional approximation space. Both methods begin from the same Bellman logic, but differ in the object being approximated and in the way approximation error is interpreted.

3 Risk-neutral model and benchmark pricing framework

The project uses a standard Black–Scholes style GBM environment under the risk-neutral measure:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}},$$

with constant volatility σ and no dividends. In discrete time,

$$S_{t+\Delta t} = S_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z\right), \quad Z \sim N(0, 1).$$

This benchmark is chosen for a practical reason. It provides a common one-dimensional testbed under which both Longstaff–Schwartz and Tsitsiklis–Van Roy can be compared against direct American pricing methods. Since the main purpose of the report is methodological comparison rather than market calibration, a shared synthetic benchmark is more useful here than a historical calibration exercise.

The shared synthetic benchmark uses the following grid:

- $K = 40$,
- $r = 0.06$,
- $S_0 \in \{36, 38, 40, 42, 44\}$,
- $\sigma \in \{0.20, 0.40\}$,
- $T \in \{1, 2\}$,
- 50 exercise dates per year,
- 100,000 simulated paths, generated as 50,000 base paths and 50,000 antithetic paths.

This testbed is used for both Longstaff and TVR whenever possible. That common design is important because it allows differences in results to be interpreted as methodological differences rather than differences in case construction.

Three benchmark layers are used in the vanilla study. The Black–Scholes European put is the lower reference,

$$P^{\text{Euro}} = Ke^{-rT}N(-d_2) - S_0N(-d_1),$$

with the standard definitions of d_1 and d_2 . It is a useful lower bound because, for a non-dividend-paying stock, the American put should not be worth less than the corresponding European put under the same model inputs. However, it is not the direct benchmark for the American claim, because it prices a different contract without early exercise.

The primary direct American benchmark is an implicit finite-difference solver. This is treated as the main benchmark because it solves the same one-dimensional American stopping problem under the same GBM assumptions. A CRR tree is also included as an auxiliary check. This layered benchmark design is consistent with the logic of the vanilla American put example in Longstaff–Schwartz: simulation output should be judged against a direct method that solves the same early-exercise problem, while the European price serves as a lower reference rather than as the benchmark itself.

It is important to distinguish these project assumptions from the assumptions of the original papers. Longstaff–Schwartz emphasize that the LSM idea is not tied to GBM and can be applied in path-dependent, multifactor, and jump settings. Tsitsiklis–Van Roy similarly formulate their method for complex American-style options with possibly high-dimensional state spaces. In this report, GBM is therefore not a restriction imposed by the papers themselves, but a common benchmark environment used to compare the two methods cleanly in the vanilla case before moving to richer extensions.

4 Longstaff–Schwartz (2001): paper walkthrough and intuition

Longstaff and Schwartz (2001) address the main difficulty in simulation-based American pricing: simulation is well suited to terminal expected values, but American-style claims require continuation values and exercise decisions at intermediate dates. Their key idea is that the conditional expected payoff from continuation can be estimated from the cross-sectional information in simulated paths by least-squares regression.

At an exercise date t_i , the continuation value is approximated by

$$C_i(S_{t_i}) \approx \hat{C}_i(S_{t_i}) = \sum_{j=0}^m \beta_{i,j} \psi_j(S_{t_i}),$$

where $\{\psi_j\}$ is a chosen basis. In the vanilla American put example, Longstaff–Schwartz use a constant together with low-order weighted Laguerre polynomials, motivated by the fact that the continuation value is a conditional expectation and can be approximated by projection onto a finite-dimensional basis.

The algorithm proceeds backward. First, simulate risk-neutral paths and initialize the option payoff at maturity. Then, moving backward through the exercise dates, regress discounted realized future cash flows on basis functions of the current state. This fitted value is used as an estimate of the continuation value. The holder exercises whenever the intrinsic value exceeds the fitted continuation value, and otherwise continues. The important point is that the regression target is not the next-step intrinsic payoff, but the discounted realized cash flow generated by continuing and then following the estimated stopping policy thereafter.

A practical feature of the method is that the regression is carried out only on in-the-money paths. The reason is that the exercise decision is economically relevant only in that region. Longstaff and Schwartz argue that restricting the regression to in-the-money states improves efficiency because the conditional expectation does not need to be approximated accurately in regions where immediate exercise is impossible anyway.

A useful theoretical interpretation of the method is that any fixed finite basis and exercise design produces a particular stopping rule. The resulting price is therefore the value of that rule rather than the unknown true American value itself. This helps explain why the method is often described as having a lower-bound logic asymptotically, but also why finite-sample implementations need not lie below every numerical benchmark in every case. Later convergence results, such as Stentoft (2004), make clear that consistency of the LSM estimator requires attention not only to the number of simulated paths but also to the richness of the approximation space.

5 Tsitsiklis–Van Roy (2001): paper walkthrough and intuition

Tsitsiklis and Van Roy (2001) study American-style option pricing as a finite-horizon optimal stopping problem solved by approximate dynamic programming. Their starting point is the discrete-time Bellman recursion, but instead of computing the value function exactly over the full state space, they work within a finitely parameterized approximation class and update the approximation recursively backward through time. This makes the method suitable for complex American-style claims with possibly high-dimensional state spaces.

In their framework, the continuation or value function is approximated by projection onto a finite-dimensional basis,

$$\hat{C}_i(s) = \sum_{j=0}^m \alpha_{i,j} \phi_j(s),$$

and the approximate value recursion is written as

$$\hat{V}_i(s) = \max\{h(s), \hat{C}_i(s)\}.$$

The key difference from Longstaff–Schwartz is that the regression target is based on discounted next-step estimated values, not discounted realized future cash flows. In this sense, TVR is closer to fitted value iteration than to direct continuation-cash-flow regression.

A central concept in the paper is the use of representative states. The regressions are carried out not over the entire state space, but at a finite sample of states, and the resulting fitted function is used to approximate values elsewhere. The paper shows that this design choice matters greatly. If representative states are chosen arbitrarily, approximation error can grow rapidly with the time horizon. By contrast, if they are generated by simulating the state process under the risk-neutral distribution, the approximation error remains controlled. This result is one of the main theoretical motivations for trajectory-based regression methods.

Another important feature of the paper is its explicit emphasis on approximation error. The relevant question is not whether one has chosen a particular named polynomial family, but whether the selected approximation space can represent the true continuation structure well over the region of the state space actually visited by simulated paths. This is why the paper frames the method in terms of projection error and recursive error propagation rather than only in terms of regression fit at a single date.

In the present project, TVR is implemented under the same shared testbeds as Longstaff–Schwartz so that differences in the results can be interpreted as differences in recursion logic rather than differences in model environment. The practical output is again a stopping policy, but the conceptual route is different: Longstaff estimates continuation from realized future payoffs, whereas TVR recursively approximates the value function itself.

6 Conceptual Comparison of the Two Papers

The two papers solve closely related American-style pricing problems, but they do so through different approximation targets. Longstaff–Schwartz estimate continuation values by regressing discounted realized future cash flows on functions of the current state and then infer a stopping rule from that comparison. Tsitsiklis–Van Roy instead use discounted next-step estimated values and frame the problem more explicitly as approximate dynamic programming. In both cases, the underlying Bellman logic is the same, but the recursion target is different.

This distinction matters for interpretation. In Longstaff–Schwartz, the regression is tied directly to the realized payoff from continuation along simulated paths, so the method is naturally read as a continuation-regression approach. In Tsitsiklis–Van Roy, the emphasis is on recursively approximating the value function over a finite-dimensional approximation space, so the method is more naturally read as fitted value iteration. The two approaches therefore differ not only computationally, but also conceptually in how they describe the role of regression in the stopping problem.

In this project, Longstaff–Schwartz serves as the main empirical pricing procedure and Tsitsiklis–Van Roy serves as a matched comparator under the same benchmark design. The comparison is therefore methodological: both methods face the same simulated states, exercise grids, payoff structures, and benchmark references, but differ in recursion logic and approximation target.

6.1 Basis Functions and Approximation Viewpoint

Both papers require approximation of a continuation-related function at each backward time step using a finite set of basis functions. Basis choice is therefore a substantive modeling decision rather than a cosmetic implementation detail. It affects approximation quality, numerical stability, and the behavior of the implied stopping rule. The two papers, however, motivate the role of basis functions somewhat differently. Longstaff–Schwartz present basis functions mainly as a practical way to estimate continuation values accurately enough for exercise decisions, whereas Tsitsiklis–Van Roy place more emphasis on projection error and on how well the chosen approximation space represents the relevant value function over the state distribution induced by the model.

6.1.1 Basis Functions in Longstaff–Schwartz (2001)

In the Longstaff–Schwartz framework, basis functions are used to approximate the conditional expectation of continuation value at each exercise date. For the vanilla American put, the paper uses a constant together with the first three weighted Laguerre polynomials in normalized spot, where

$$X = \frac{S_t}{K}.$$

The first three weighted Laguerre terms are

$$\begin{aligned} L_0(X) &= e^{-X/2}, \\ L_1(X) &= e^{-X/2}(1 - X), \\ L_2(X) &= e^{-X/2} \left(1 - 2X + \frac{X^2}{2} \right). \end{aligned}$$

The continuation function is then approximated by a linear combination of basis terms,

$$F(\omega; t_k) \approx \sum_{j=0}^M a_j \psi_j(X),$$

with coefficients estimated by least squares on the relevant simulated paths. In the vanilla benchmark, this basis is applied to in-the-money paths because those are the states for which the exercise decision is economically active.

For higher-dimensional examples, Longstaff and Schwartz expand the basis to reflect the richer state vector. In the American–Bermuda–Asian setting, for example, the continuation value depends not only on current spot but also on the running average, so the basis must incorporate functions of both state variables and their interactions. This is directly relevant to the project’s Asian-style extension, where the move from a one-state vanilla benchmark to a two-state continuation problem is one of the main methodological motivations.

A useful empirical point in Longstaff–Schwartz is that pricing accuracy in simple settings is often not very sensitive to the precise orthogonal family used, provided the basis is rich enough to approximate the continuation function reasonably well. The paper reports that Laguerre polynomials, ordinary powers, Hermite polynomials, and trigonometric functions can all give very similar vanilla-put prices. The practical lesson is not that basis choice is irrelevant, but that in smooth low-dimensional problems several reasonable low-order bases may work similarly well. At the same time, the paper warns that poor scaling and near-singular regression matrices can still create numerical instability, which is why normalizing the state variable by strike is a sensible implementation choice.

6.1.2 Basis Functions in Tsitsiklis–Van Roy (2001)

In the Tsitsiklis–Van Roy framework, basis functions play a closely related but more explicitly theoretical role. Rather than directly approximating realized continuation cash flows, the method parameterizes the continuation or value function as a linear combination of basis elements,

$$\tilde{Q}(x, r) = \sum_{k=1}^K r_k \phi_k(x) = (\Phi r)(x),$$

where $\phi_1, \dots, \phi_K$ are fixed basis functions and $r \in \mathbb{R}^K$ is the regression parameter vector. What matters in the TVR analysis is not a particular polynomial family by name, but how well the chosen finite-dimensional space can approximate the true continuation function under the relevant state distribution.

This idea is summarized by the best approximation error

$$\epsilon_n^* = \min_r \|\Phi r - Q_n\|_{\pi_n} = \|\Pi_n Q_n - Q_n\|_{\pi_n},$$

where π_n is the distribution of representative states at time step n . In the TVR framework, the quality of the algorithm depends on keeping this approximation error controlled through the backward recursion. The theoretical emphasis is therefore less on a named basis family and more on whether the chosen basis captures the relevant continuation structure over the state region visited by simulated paths.

This perspective is useful for the present project because it clarifies why basis design should be treated as a substantive modeling choice rather than a cosmetic implementation detail. In the shared vanilla benchmark, both Longstaff and TVR are intentionally run with matched approximation complexity so that the comparison is not driven by one method being given a systematically richer function class than the other. In the Asian-style extension, both methods move from a one-state to a two-state continuation problem, and the basis must expand accordingly to reflect the richer state vector. The practical question is therefore not whether one paper prescribes a universally superior polynomial family, but whether the chosen basis is rich enough to approximate continuation well while remaining numerically stable under the simulated trajectory distribution.

Overall, the two papers offer complementary viewpoints on basis selection. Longstaff–Schwartz shows empirically that several reasonable low-order bases can perform similarly in simple problems, while Tsitsiklis–Van Roy explains theoretically that what ultimately matters is the projection error induced by the chosen approximation space. Read together, they support the design logic of the present study: use a shared benchmark environment, keep basis complexity comparable across methods, and then examine through robustness analysis how pricing performance reacts to basis choice, validation design, and state-space complexity.

7 Experimental Design

The project is organized around three shared study layers.

The first layer is the vanilla American put benchmark under risk-neutral GBM, which serves as the main validation exercise of the project. In the one-dimensional vanilla-put setting, the implicit finite-difference solver is treated as the main direct American benchmark, consistent with the validation style of Longstaff and Schwartz, who compare their simulation values with finite-difference American prices and report the corresponding European values from the closed-form Black–Scholes formula. The Black–Scholes European put is therefore used here as a lower reference for the corresponding European contract, not as a benchmark for the American claim itself. A CRR tree is also reported, but only as an auxiliary American consistency

check rather than as the primary benchmark. Since Tsitsiklis and Van Roy do not organize their study around the same benchmark table structure as Longstaff and Schwartz, this project imposes a common benchmark framework for comparison: both Longstaff and TVR are run on the same synthetic grid, under the same exercise-date conventions and path counts, and are evaluated against the same finite-difference American benchmark, with Black-Scholes and CRR included as supporting references. This benchmark stack allows the comparison to say more than simply whether one regression method prices above or below another; it makes it possible to assess whether the regression-based prices are reasonable relative to direct American solution methods under the same model assumptions.

The second layer is an exotic American-Bermuda-Asian option extension, motivated by the second Longstaff-Schwartz example, which shows why regression-based Monte Carlo becomes attractive once the payoff is path dependent and the state is no longer one-dimensional. In our implementation, the state includes current spot and the running arithmetic average, exercise is prohibited until a lockout date, and the payoff depends on the average rather than on spot alone. This is the key design motive of the extension: to move from the vanilla benchmark, where trees and finite differences remain feasible, to a setting in which continuation values depend on multiple state variables and the comparative advantage of regression-based methods becomes more evident. The purpose is therefore not to construct a second direct benchmark exercise of the same type as the vanilla put, but to study how both Longstaff and TVR behave once the state vector is richer and the continuation regression becomes genuinely multivariate.

The third layer is an American option on asset which follows a jump-diffusion process, motivated by the fourth Longstaff-Schwartz example, which illustrates that the least-squares Monte Carlo framework is not tied to standard Black-Scholes dynamics and can be applied when the underlying follows a process with jumps. In our implementation, the standard GBM simulator is replaced by a stylized jump-to-ruin process, while the pricing workflow within each method remains unchanged. The design motive here is to isolate model flexibility: the state dynamics are altered, but the regression-and-backward-induction architecture is left intact. This makes the jump case a useful structural test of both Longstaff and TVR under non-GBM dynamics. As with the Asian-style extension, the point is not to provide the same kind of direct one-dimensional benchmark used in the vanilla study, but to demonstrate that both regression frameworks extend naturally beyond the standard GBM benchmark and to connect that extension explicitly to the broader message of the original papers.

Robustness analysis is conducted on the vanilla benchmark because that is where a direct American reference is available. On the Longstaff side, the study includes in-sample versus out-of-sample valuation, path sensitivity, exercise-grid sensitivity, basis sensitivity, and seed sensitivity. On the TVR side, the study includes in-sample versus out-of-sample valuation and compact path, timestep, and basis sensitivity exercises. The role of these diagnostics is interpretive as much as numerical: they show how the methods respond to design choices in basis specification, exercise grid, and validation structure. In particular, they help distinguish whether pricing differences come mainly from Monte Carlo noise, from approximation design, or from the backward-recursion logic itself.

A clean way to summarize the experimental hierarchy in the report is:

European BS lower reference $\longrightarrow$ Direct American benchmarks (FD, CRR) $\longrightarrow$
Regression-based methods (Longstaff, TVR) $\longrightarrow$ Robustness and extensions.

The regression methods are being judged against a benchmark layer that matches the vanilla American option problem as closely as possible.

8 Integrated Implementation Workflow and Code Design

The implementation is organized as one shared pricing study with two regression engines. Basis construction, simulation, and direct benchmarking are shared as far as possible. What changes between the two frameworks is the backward recursion itself. Both Longstaff and TVR are run on the same synthetic vanilla American put grid so that differences in results can be interpreted as methodological differences rather than differences in case construction.

At the basis level, the project uses matched approximation complexity across methods. The vanilla benchmark uses a one-state basis built from weighted Laguerre terms in normalized spot. The Asian-style extension uses a two-state basis in current spot and running arithmetic average, together with selected interaction terms. This matching matters because the comparison should not be driven by one method having a systematically richer approximation space than the other.

At the simulation level, both methods use the same model-side path generators whenever possible. The vanilla study uses GBM with antithetic variates. The Asian-style extension uses the same spot process augmented by a running arithmetic average. The jump-style extension uses a jump-to-ruin process. This shared simulator structure is important because it isolates the effect of the recursion method from the effect of the state process.

The Longstaff engine implements backward induction by regressing discounted realized continuation cash flows on in-the-money paths. The TVR engine implements fitted value iteration by regressing discounted next-step estimated values on current-state basis functions. In both cases, the fitted rule is turned into an implied stopping policy and then evaluated on a fresh path set. That shared policy-evaluation step makes the comparison much more informative than a comparison of fitted time-zero values alone.

8.1 Benchmark Layer: what the reference methods are and why they are used

The benchmark layer provides the common pricing references in the vanilla study: the Black–Scholes European put, the implicit finite-difference American put, and the Cox–Ross–Rubinstein (CRR) American tree. These are not included as competing research contributions; they are included because they anchor the regression-based methods against standard one-dimensional benchmarks under the same GBM assumptions. Longstaff’s own vanilla benchmark is framed exactly this way: simulation values are checked against finite-difference values, while the European put is given by the closed-form Black–Scholes formula.

8.1.1 Black–Scholes European Benchmark

The Black–Scholes price is the value of the **European** put under the same risk-neutral GBM model used in the simulation benchmark. It is a useful lower reference because it prices the same underlying and strike under the same model inputs, but without early exercise. For a non-dividend-paying stock, the American put should not be worth less than the European put, so the Black–Scholes value is an immediate sanity check. It is **not** the direct benchmark for the American claim, because it prices a different contract.

For spot (S_0), strike (K), volatility (σ), risk-free rate (r), and maturity (T), the European put price is

$$P_{\text{Euro}} = Ke^{-rT}N(-d_2) - S_0N(-d_1),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

In the report, this benchmark should be interpreted as a lower reference and a way to quantify the early-exercise premium:

$$\text{Early-exercise premium} = P_{\text{American}} - P_{\text{Euro}}.$$

8.1.2 Implicit Finite-difference American Benchmark

The finite-difference method is the main direct American benchmark in the vanilla study because it solves the same one-dimensional early-exercise problem under the same GBM model. Instead of simulating paths, it works on a grid in stock price and time and solves the Black–Scholes partial differential equation backward subject to the American early-exercise constraint. This makes it much more appropriate than Black–Scholes for validating the vanilla American put.

Under risk-neutral GBM, the American put value ($V(S, t)$) satisfies the Black–Scholes PDE in the continuation region:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

At maturity,

$$V(S, T) = (K - S)^+.$$

For an American put, the value must also satisfy the early-exercise inequality

$$V(S, t) \geq (K - S)^+,$$

so backward induction is carried out with projection onto intrinsic value at each time level. In an implicit finite-difference scheme, the time derivative is handled backward in time and the spatial derivatives are discretized on an (S)-grid, producing a linear system at each step. After solving that system, the numerical value is replaced by

$$V_i^n = \max(V_i^{n, \text{cont}}, K - S_i),$$

where ($V_i^{n, \text{cont}}$) is the continuation-region finite-difference update at node (S_i, t_n).

8.1.3 CRR Binomial Tree Benchmark

The CRR tree is included as an auxiliary American benchmark and an additional consistency check. Like finite differences, it solves the discrete-time early-exercise problem directly by backward induction, but it does so on a recombining lattice rather than a continuous stock-price grid.

Let ($\Delta t = T/N$). In the CRR tree, the stock moves each step by

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u},$$

and the risk-neutral probability is

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

At maturity, the option value is

$$V_N(j) = (K - S_N(j))^+.$$

Then the tree is rolled backward using

$$C_n(j) = e^{-r\Delta t}(pV_{n+1}(j+1) + (1-p)V_{n+1}(j)),$$

and the American value at each node is

$$V_n(j) = \max((K - S_n(j))^+, C_n(j)).$$

In this project, the CRR tree is not the primary benchmark because the finite-difference solver already plays that role. Instead, it serves as a supporting check that the direct American benchmark layer is internally coherent.

8.2 Overall implementation logic

Putting these pieces together, the workflow is as follows. First, choose the shared model inputs (S_0, K, r, σ, T) under the synthetic GBM benchmark. Second, compute the benchmark layer: Black–Scholes for the European lower reference, finite differences for the main direct American benchmark, and CRR as an auxiliary American check. Third, run the two regression-based methods under the same simulated-path design. Fourth, compare Longstaff and TVR not just to each other, but to the benchmark layer that defines what a plausible vanilla American price should look like. This is why the vanilla scripts for Longstaff and TVR operate under the same benchmark philosophy, and the Asian-style and jump-style scripts operate under the same extension philosophy. The result is a single workflow with shared testbeds and different recursion logic rather than two disconnected coding exercises.

9 Results

9.1 Shared Vanilla Benchmark

spot	sigma	maturity	American Put FD	American Put LSMC	LSMCE SE	Fit-Eval	LSMC Premium - FD Premium
36.0	0.2	1	4.463500	4.205800	0.012700	0.246600	-0.257700
38.0	0.2	1	3.235000	3.266800	0.011200	0.184000	0.031800
40.0	0.2	1	2.299500	2.443600	0.009900	0.134100	0.144100
42.0	0.2	1	1.604600	1.745600	0.008500	0.097800	0.140900
44.0	0.2	1	1.099800	1.194600	0.007100	0.067000	0.094700
36.0	0.2	2	4.828000	4.157000	0.014600	0.499500	-0.671000
38.0	0.2	2	3.732800	3.415500	0.013200	0.403200	-0.317400
40.0	0.2	2	2.873400	2.727100	0.011900	0.318700	-0.146300
42.0	0.2	2	2.202400	2.138600	0.010700	0.254600	-0.063900
44.0	0.2	2	1.681000	1.656100	0.009600	0.199500	-0.024900
36.0	0.4	1	7.076000	7.219800	0.021200	0.415600	0.143900
38.0	0.4	1	6.121600	6.368100	0.020200	0.361600	0.246500
40.0	0.4	1	5.286100	5.487900	0.019200	0.305700	0.201800
42.0	0.4	1	4.557600	4.783900	0.018300	0.269900	0.226200
44.0	0.4	1	3.924000	4.132100	0.017200	0.231100	0.208100
36.0	0.4	2	8.486800	8.351000	0.025000	1.020800	-0.135800
38.0	0.4	2	7.647800	7.652800	0.024200	0.924000	0.005000
40.0	0.4	2	6.896900	6.971400	0.023500	0.852600	0.074500
42.0	0.4	2	6.224600	6.255400	0.022800	0.749500	0.030800
44.0	0.4	2	5.621900	5.666500	0.022100	0.678800	0.044600

Table 1: Vanilla American put pricing comparison under different spot, volatility, and maturity settings.

Table 1 shows that the revised Longstaff benchmark produces American put values of the correct order of magnitude across the full parameter grid. The reported two-pass values are generally closer to the finite-difference benchmark than the original same-path values, but the gap does not disappear uniformly. In some cases the method remains above the benchmark, while in others it falls below it. The column recording the fit-versus-evaluation gap is particularly informative because it quantifies the effect of same-path optimism in the original implementation.

Table 2: Tsitsiklis–Van Roy vanilla benchmark under the same synthetic t

	spot	sigma	maturity	american_put_fd	tvr_price_in_sample	tvr_price_out_of_sample	tvr_stderr
0	36.0	0.2	1.0	4.4635	4.3861	4.3851	0.0114
1	38.0	0.2	1.0	3.2350	3.1977	3.1925	0.0106
2	40.0	0.2	1.0	2.2995	2.2864	2.2783	0.0094
3	42.0	0.2	1.0	1.6046	1.5999	1.5957	0.0082
4	44.0	0.2	1.0	1.0998	1.0966	1.0912	0.0070
5	36.0	0.2	2.0	4.8280	4.6962	4.7065	0.0135
6	38.0	0.2	2.0	3.7328	3.6604	3.6598	0.0127
7	40.0	0.2	2.0	2.8734	2.8240	2.8285	0.0118
8	42.0	0.2	2.0	2.2024	2.1582	2.1582	0.0107
9	44.0	0.2	2.0	1.6810	1.6368	1.6378	0.0097
10	36.0	0.4	1.0	7.0760	6.9015	6.9009	0.0216
11	38.0	0.4	1.0	6.1216	5.9237	5.9223	0.0210
12	40.0	0.4	1.0	5.2861	5.1318	5.1268	0.0200
13	42.0	0.4	1.0	4.5576	4.4370	4.4366	0.0189
14	44.0	0.4	1.0	3.9240	3.8354	3.8380	0.0178
15	36.0	0.4	2.0	8.4868	8.4146	8.4300	0.0247
16	38.0	0.4	2.0	7.6478	7.5979	7.6128	0.0242
17	40.0	0.4	2.0	6.8969	6.8597	6.8591	0.0236
18	42.0	0.4	2.0	6.2246	6.2001	6.1972	0.0230
19	44.0	0.4	2.0	5.6219	5.5991	5.5954	0.0223

Table 2 places TVR under the same benchmark conditions. The out-of-sample TVR values are generally close to the finite-difference price and are often slightly below it. This is a useful result because it shows that the revised TVR implementation is no longer behaving like the older loose prototype that materially overstated vanilla prices. The final comparison is therefore between two benchmarked regression procedures rather than between one benchmarked method and one poorly aligned comparator.

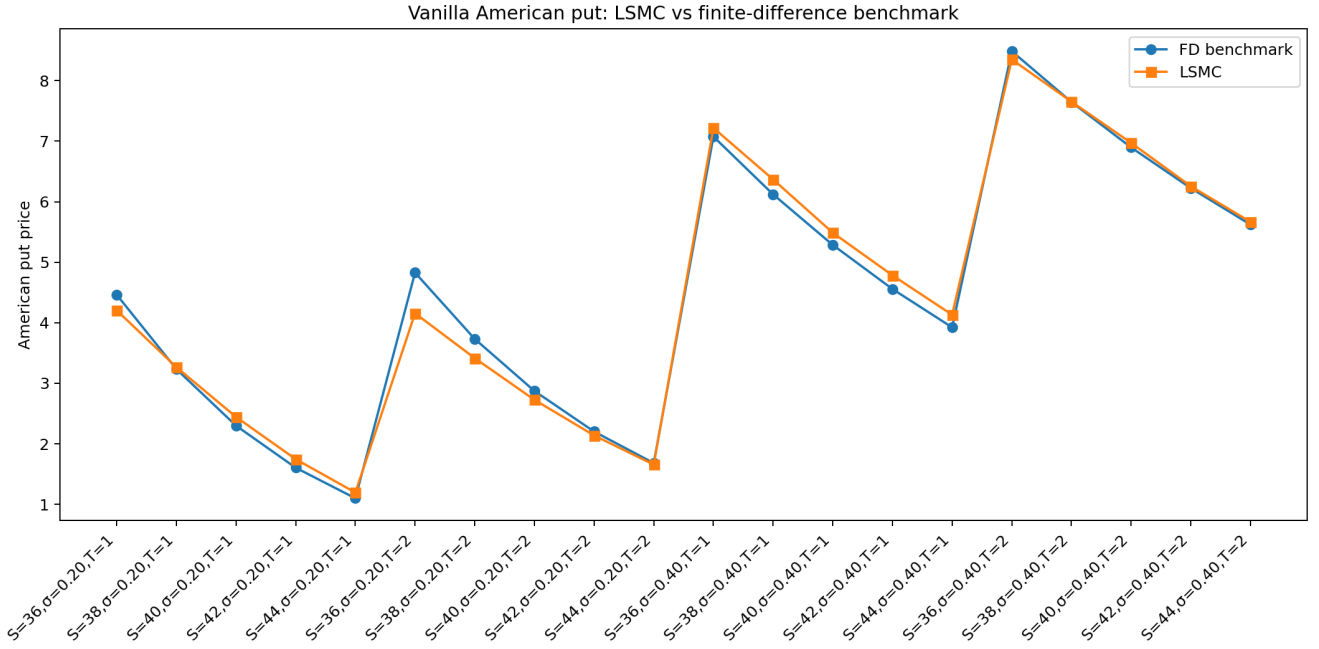


Figure 1: Vanilla benchmark comparison: Longstaffe LSMC versus finite-difference benchmark.

Figure 1 shows the Longstaffe side of the shared vanilla benchmark. The visual pattern matches the table: the revised benchmark is much tighter than the earlier same-path version, but there is still visible case-by-case variation around the finite-difference reference. That variation is part of the substantive numerical story of the method rather than a formatting artifact.

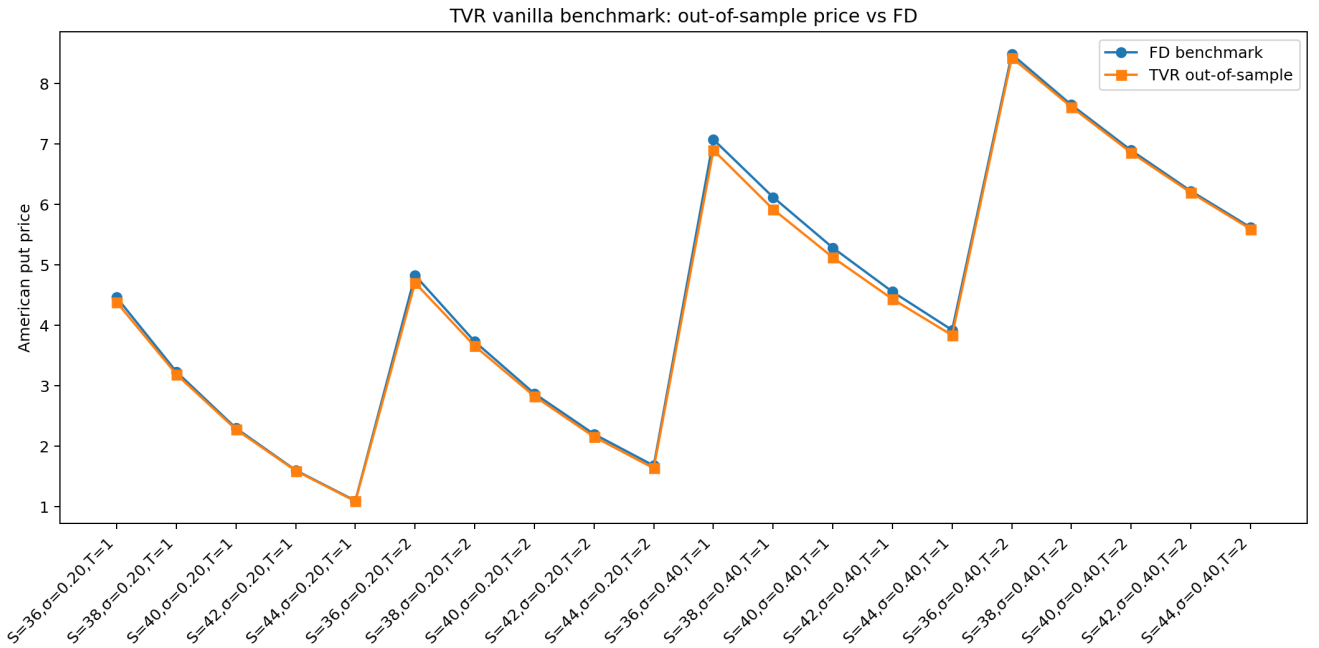


Figure 2: Vanilla benchmark comparison: TVR out-of-sample value versus finite-difference benchmark.

Figure 2 shows the corresponding TVR benchmark. The TVR values cluster closely around the finite-difference benchmark across the shared grid. Read together with Table 2, the figure suggests that the

revised TVR comparator is reasonably well aligned to the common testbed and can be interpreted as a genuine methodological alternative rather than a separate ad hoc calculation.

9.2 Shared Exercise-policy Interpretation

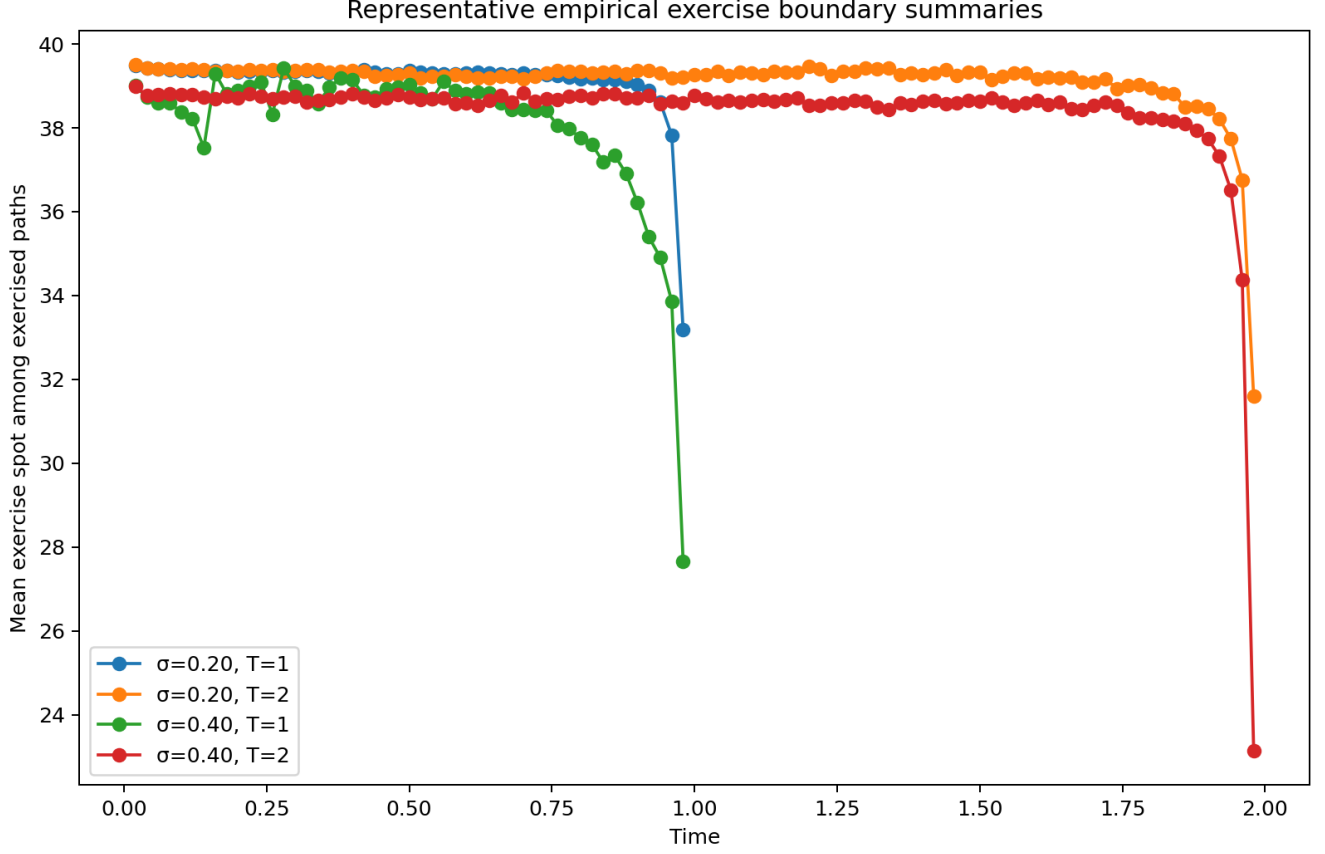


Figure 3: Empirical exercise-boundary summary: Longstaff vanilla benchmark.

Figure 3 summarizes exercise behavior for representative Longstaff vanilla cases. It should not be read as an exact free boundary. Rather, it is an empirical summary of states at which exercise occurs on simulated paths. The figure is still useful because it shows where the estimated stopping rule becomes active and how that region shifts across maturity and volatility settings.

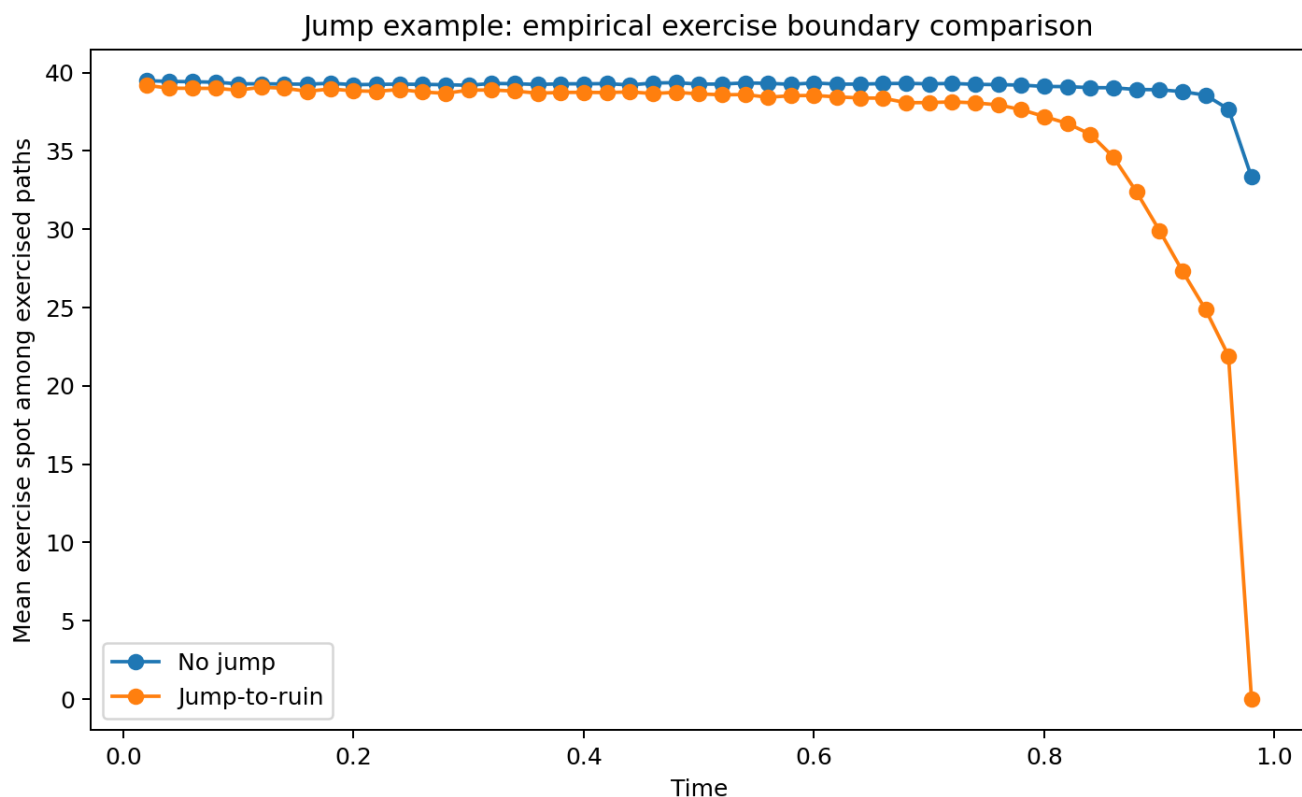


Figure 4: Jump-style boundary comparison: Longstaff.

The Longstaff jump-style boundary figure extends the same idea to the jump-to-ruin setting. Once downward jump risk is introduced, exercise decisions occur under very different state dynamics. The figure should again be read as a summary of simulated exercise behavior rather than as a structural boundary, but it makes the change in stopping behavior visible.

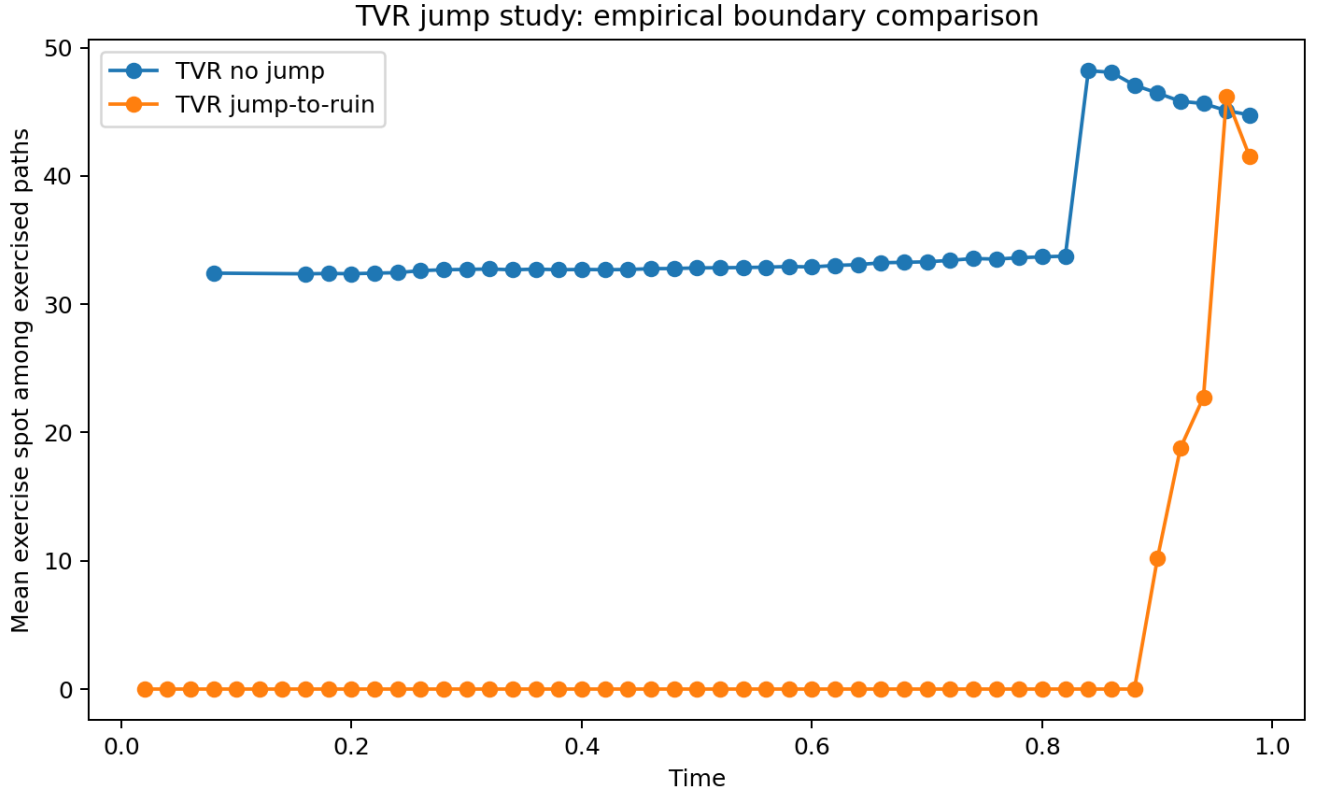


Figure 5: Jump-style boundary comparison: TVR.

Figure 5 provides the analogous TVR boundary summary. The qualitative pattern is similar: exercise tends to occur in lower stock-price regions, and the jump case alters the timing and location of exercise. The key methodological point is that both engines generate interpretable stopping behavior under the same changed dynamics.

9.3 Shared Robustness Analysis

Table 3: Longstaff in-sample versus out-of-sample policy value

	spot	sigma	maturity	fit_seed	eval_seed	in_sample_price	in_sample_stderr	out_of_sample_price
0	36.0	0.2	1.0	20260406	20260406	4.4524	0.0135	4.2057
1	36.0	0.2	1.0	20260407	20260407	4.4517	0.0135	4.2050
2	36.0	0.2	1.0	20260408	20260408	4.4519	0.0135	4.2052
3	40.0	0.2	2.0	20261406	20262406	3.0596	0.0134	2.7442
4	40.0	0.2	2.0	20261407	20262407	3.0275	0.0133	2.7219
5	40.0	0.2	2.0	20261408	20262408	3.0201	0.0134	2.6921
6	44.0	0.4	1.0	20262406	20264406	4.3604	0.0183	4.1103
7	44.0	0.4	1.0	20262407	20264407	4.3908	0.0184	4.1526
8	44.0	0.4	1.0	20262408	20264408	4.3708	0.0183	4.1197
9	40.0	0.4	2.0	20263406	20266406	7.8097	0.0265	6.9652
10	40.0	0.4	2.0	20263407	20266407	7.6596	0.0265	6.8219
11	40.0	0.4	2.0	20263408	20266408	7.8020	0.0265	6.9743

Table 4: TVR in-sample versus out-of-sample policy valuation for representative vanilla cases.

	spot	sigma	maturity	fit_seed	eval_seed	in_sample_price	out_of_sample_price	difference	stderr
0	36.0	0.2	1.0	20260407	20265407	4.3861	4.3810	0.0050	0.0114
1	36.0	0.2	1.0	20260408	20265408	4.4016	4.3894	0.0122	0.0114
2	36.0	0.2	1.0	20260409	20265409	4.3960	4.3882	0.0078	0.0113
3	40.0	0.2	2.0	20261407	20267407	2.8035	2.8173	-0.0138	0.0117
4	40.0	0.2	2.0	20261408	20267408	2.8148	2.8109	0.0039	0.0118
5	40.0	0.2	2.0	20261409	20267409	2.8219	2.8296	-0.0077	0.0118
6	44.0	0.4	1.0	20262407	20269407	3.8648	3.8395	0.0253	0.0178
7	44.0	0.4	1.0	20262408	20269408	3.8555	3.8601	-0.0046	0.0178
8	44.0	0.4	1.0	20262409	20269409	3.8523	3.8571	-0.0048	0.0178
9	40.0	0.4	2.0	20263407	20271407	6.8432	6.8289	0.0143	0.0236
10	40.0	0.4	2.0	20263408	20271408	6.8294	6.8507	-0.0214	0.0236
11	40.0	0.4	2.0	20263409	20271409	6.8481	6.8600	-0.0119	0.0237

Table 3 and Table 4 report the central fit-versus-evaluation diagnostic for the two methods. On the Longstaff side, the gap between in-sample and out-of-sample values is clearly visible and becomes substantial in the harder high-volatility, long-maturity case. On the TVR side, the same diagnostic is much tighter in the representative cases shown here. The contrast does not imply that TVR is categorically better; it shows that the two recursion schemes react differently to the same validation design.

Table 5: Longstaff path-count sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	paths_total	lsmc_price	lsmc_stderr	fd_benchmark	lsmc_minus_fd
0	40.0	0.2	1.0	25000	2.3174	0.0194	2.2995	0.0179
1	40.0	0.2	1.0	50000	2.3878	0.0139	2.2995	0.0884
2	40.0	0.2	1.0	100000	2.4472	0.0099	2.2995	0.1477
3	40.0	0.2	1.0	200000	2.4474	0.0070	2.2995	0.1479
4	40.0	0.4	2.0	25000	6.9352	0.0467	6.8969	0.0383
5	40.0	0.4	2.0	50000	6.8354	0.0333	6.8969	-0.0615
6	40.0	0.4	2.0	100000	6.8574	0.0235	6.8969	-0.0395
7	40.0	0.4	2.0	200000	6.9576	0.0166	6.8969	0.0607

Table 6: TVR path-count sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	paths_total	tvr_price_out_of_sample	tvr_stderr_out_of_sample	fd_benchmark
0	40.0	0.2	1.0	25000	2.2703	0.0191	2.2995
1	40.0	0.2	1.0	50000	2.2990	0.0134	2.2995
2	40.0	0.2	1.0	100000	2.2909	0.0094	2.2995
3	40.0	0.2	1.0	200000	2.2818	0.0067	2.2995
4	40.0	0.4	2.0	25000	6.8895	0.0472	6.8969
5	40.0	0.4	2.0	50000	6.8606	0.0334	6.8969
6	40.0	0.4	2.0	100000	6.8318	0.0236	6.8969
7	40.0	0.4	2.0	200000	6.8500	0.0167	6.8969

Table 5 and Table 6 show that increasing path count reduces standard error, but does not by itself guarantee benchmark convergence. The Longstaff values move around the finite-difference benchmark

in a case-dependent way, while TVR stays relatively close to it across the tested path counts. This comparison reinforces the idea that path count is only one element of performance; recursion target and policy construction matter as well.

Table 7: Longstaff exercise-grid sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	exercise_dates_per_year	lsmc_price	lsmc_stderr	fd_benchmark	lsmc_minus
0	40.0	0.2	1.0	25	2.5610	0.0097	2.2821	0.2789
1	40.0	0.2	1.0	50	2.4399	0.0098	2.2995	0.1404
2	40.0	0.2	1.0	100	2.3439	0.0100	2.3086	0.0353
3	40.0	0.4	2.0	25	7.2098	0.0230	6.8724	0.3373
4	40.0	0.4	2.0	50	6.8366	0.0235	6.8969	-0.0603
5	40.0	0.4	2.0	100	6.8272	0.0238	6.9096	-0.0824

Table 8: TVR exercise-grid sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	exercise_dates_per_year	tvr_price_out_of_sample	tvr_stderr_out_of_sample
0	40.0	0.2	1.0	25	2.2954	0.0093
1	40.0	0.2	1.0	50	2.2906	0.0094
2	40.0	0.2	1.0	100	2.2809	0.0095
3	40.0	0.4	2.0	25	6.8635	0.0233
4	40.0	0.4	2.0	50	6.8611	0.0236
5	40.0	0.4	2.0	100	6.8189	0.0239

Table 7 and Table 8 show that both methods are sensitive to the exercise grid. Longstaff can shift from benchmark overvaluation to benchmark undervaluation as the number of exercise dates changes. TVR remains comparatively close to the benchmark in the selected cases, but the effect of the grid is still visible. The main lesson is that discretization is part of the modeling design, not a neutral background detail.

Table 9: Longstaff basis-size sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	basis_size	lsmc_price	lsmc_stderr	fd_benchmark	lsmc_minus_fd
0	40.0	0.2	1.0	2	2.3846	0.0099	2.2995	0.0851
1	40.0	0.2	1.0	3	2.4432	0.0099	2.2995	0.1438
2	40.0	0.2	1.0	4	2.4327	0.0099	2.2995	0.1332
3	40.0	0.2	1.0	5	2.4576	0.0098	2.2995	0.1582
4	40.0	0.4	2.0	2	6.8897	0.0235	6.8969	-0.0072
5	40.0	0.4	2.0	3	7.0039	0.0236	6.8969	0.1070
6	40.0	0.4	2.0	4	6.9752	0.0234	6.8969	0.0783
7	40.0	0.4	2.0	5	6.9922	0.0234	6.8969	0.0953

Table 10: TVR basis-size sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	basis_size	tvr_price_out_of_sample	tvr_stderr_out_of_sample	fd_benchmark
0	40.0	0.2	1.0	2	2.1510	0.0105	2.2995
1	40.0	0.2	1.0	3	2.2576	0.0099	2.2995
2	40.0	0.2	1.0	4	2.2885	0.0094	2.2995

Table 10: TVR basis-size sensitivity under the shared vanilla benchmark.

	spot	sigma	maturity	basis_size	tvr_price_out_of_sample	tvr_stderr_out_of_sample	fd_benchmark
3	40.0	0.2	1.0	5	2.2850	0.0096	2.2995
4	40.0	0.4	2.0	2	6.4664	0.0253	6.8969
5	40.0	0.4	2.0	3	6.7995	0.0244	6.8969
6	40.0	0.4	2.0	4	6.8352	0.0236	6.8969
7	40.0	0.4	2.0	5	6.8621	0.0233	6.8969

Table 9 and Table 10 show that accuracy is not monotone in the number of basis functions. A richer basis can improve approximation in principle, but it can also amplify instability in finite samples. This pattern appears for both methods, though the strength of the effect differs. The comparison supports the broader point that basis design is a substantive modeling choice rather than a purely technical parameter.

Table 11: Longstaff seed-level results for representative vanilla benchmark cases.

	spot	sigma	maturity	seed	lsmc_price	lsmc_stderr	fd_benchmark	bias_vs_fd
0	36.0	0.2	1.0	20260406	4.2058	0.0127	4.4635	-0.2577
1	36.0	0.2	1.0	20260407	4.1992	0.0127	4.4635	-0.2643
2	36.0	0.2	1.0	20260408	4.2122	0.0127	4.4635	-0.2513
3	36.0	0.2	1.0	20260409	4.2014	0.0127	4.4635	-0.2621
4	36.0	0.2	1.0	20260410	4.2201	0.0127	4.4635	-0.2433
6	40.0	0.2	2.0	20260406	2.7271	0.0119	2.8734	-0.1463
7	40.0	0.2	2.0	20260407	2.7481	0.0119	2.8734	-0.1253
8	40.0	0.2	2.0	20260408	2.7541	0.0119	2.8734	-0.1193
9	40.0	0.2	2.0	20260409	2.7243	0.0119	2.8734	-0.1491
10	40.0	0.2	2.0	20260410	2.7545	0.0120	2.8734	-0.1188
12	44.0	0.4	1.0	20260406	4.1321	0.0172	3.9240	0.2081
13	44.0	0.4	1.0	20260407	4.1841	0.0173	3.9240	0.2601
14	44.0	0.4	1.0	20260408	4.1744	0.0173	3.9240	0.2504
15	44.0	0.4	1.0	20260409	4.1210	0.0171	3.9240	0.1970
16	44.0	0.4	1.0	20260410	4.1425	0.0172	3.9240	0.2185

Table 12: Longstaff summary across repeated seeds for representative vanilla benchmark cases.

	case	mean_lsm_price	sd_across_runs	fd_benchmark	average_bias_vs_fd
5	S=36.0, =0.2, T=1.0	4.2077	0.0085	4.4635	-0.2557
11	S=40.0, =0.2, T=2.0	2.7416	0.0148	2.8734	-0.1318
17	S=44.0, =0.4, T=1.0	4.1508	0.0273	3.9240	0.2268

Table 11 and Table 12 separate the repeated-seed output into seed-level runs and case-level summaries. This avoids mixing literal summary rows into the same seed column as if they were additional replications. The seed-level table shows the dispersion across runs, while the summary table reports the mean LSM price, the standard deviation across runs, and the average bias relative to the finite-difference benchmark. Together, they make clear that Monte Carlo variation remains present even after the larger workflow revisions.

9.4 Shared Asian-style Extension

Table 1

	example	spot0	strike	risk_free	sigma	maturity	lockout_time	exercise_dates	paths
0	American-Bermuda-Asian	40.0	40.0	0.06	0.2	2.0	0.25	100	10000

Table 13 shows the Longstaff Asian-style extension under the shared design. The extension introduces a second state variable—the running arithmetic average—and a lockout period before exercise is allowed. The resulting price is therefore not directly comparable to the vanilla benchmark, but it demonstrates that the continuation regression can accommodate a richer state description within the same backward-induction framework.

	example	spot0	strike	risk_free	sigma	maturity	lockout_time	exercise_dates	pa
0	TVR Asian shared extension	40.0	40.0	0.06	0.2	2.0	0.25	100	10

Table 14 presents the corresponding TVR result. The out-of-sample TVR price exceeds its European-style Monte Carlo reference by a visible margin, indicating that the approximate dynamic-programming recursion is also capturing an early-exercise premium in the multi-state case. As on the Longstaff side, this is a compact shared extension study rather than a direct PDE-benchmarked replication.



Figure 6: Asian-style state cloud with exercise overlay: Longstaff.

Figure 6 shows the Longstaff state cloud in spot and running average, together with the exercise overlay. The figure is useful because it makes the two-state nature of the exercise problem visible. Exercise is no longer a function of spot alone; it depends on the interaction between current level and path history through the running average.

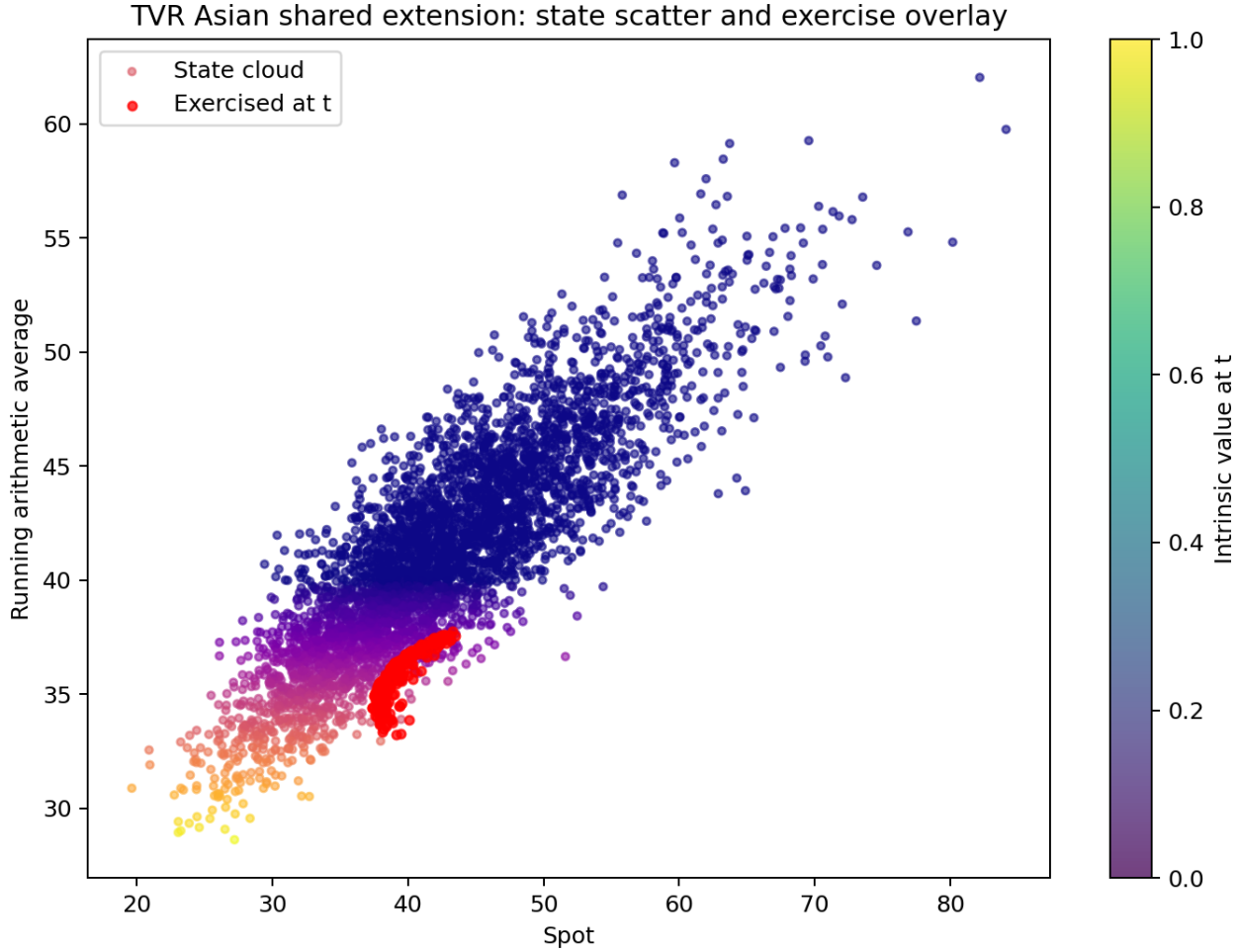


Figure 7: Asian-style state cloud with exercise overlay: TVR.

Figure 7 provides the analogous TVR visualization. The structure is similar, although the implied exercise pattern is not identical. Read together, the two figures show that both methods can operate on the same richer state space while retaining different recursive logic.

9.5 Shared Jump-style Extension

Table 15: Jump-style extension: Longstaff results.

	model	spot0	strike	sigma	risk_free	maturity	jump_intensity	american_lsm	lsm_stderr	
0	no_jump_gbm	40.0	40.0	0.2	0.06	1.0	0.0	2.5657	0.0104	2
1	jump_to_ruin	40.0	40.0	0.2	0.06	1.0	0.4	14.9459	0.0556	1

Table 15 shows the Longstaff jump-style extension. Once jump-to-ruin risk is introduced, both American and European put values rise sharply relative to the no-jump case because the stock can collapse to zero before maturity. The early-exercise premium remains positive, but its scale differs from the vanilla benchmark because the underlying risk environment has changed.

Table 16: Jump-style extension: T

	model	spot0	strike	sigma	risk_free	maturity	jump_intensity	tvr_price_in_sample	tvr_p
0	no_jump_gbm	40.0	40.0	0.2	0.06	1.0	0.0	2.2778	2.2895
1	jump_to_ruin	40.0	40.0	0.2	0.06	1.0	0.4	14.1417	14.012

Table 16 shows the same qualitative effect under TVR. The out-of-sample jump-to-ruin value is far above the no-jump value, and the early-exercise premium remains positive. The two methods therefore agree on the broad economic effect of introducing catastrophic downward jump risk, even though their recursive approximations differ.

The boundary figures in Figure 4 and Figure 5 support the same reading visually. In each case, the pricing workflow remains intact while only the simulator changes. This is the main point of the jump-style study: it demonstrates modularity with respect to the state dynamics.

9.6 Direct Longstaff vs TVR comparison

After the shared evidence has been laid out, the comparison between Longstaff and TVR can be summarized in four points. First, the recursion target differs: Longstaff regresses discounted realized future cash flows, whereas TVR regresses discounted next-step estimated values. Second, the benchmark behavior differs: in the revised vanilla study, Longstaff may sit above or below the finite-difference benchmark depending on the case, while TVR tends to remain close to it and is often slightly below it. Third, the validation behavior differs: Longstaff shows visible same-path optimism unless the policy is evaluated out of sample, while TVR’s matched implementation shows a smaller in-sample versus out-of-sample gap in the representative cases examined here. Fourth, the interpretation differs: Longstaff is more directly tied to continuation regression on in-the-money paths, while TVR more clearly reflects approximate dynamic programming through recursive value approximation.

10 Implementation and Testing

The implementation and testing logic is driven by the course rubric rather than by code structure alone. Two pricing frameworks are implemented under a common benchmark philosophy: Longstaff–Schwartz as the main practical algorithm and Tsitsiklis–Van Roy as a matched approximate-dynamic-programming comparator. Both methods are tested on the same synthetic vanilla benchmark, the same Asian-style extension, and the same jump-style extension. The purpose of this design is not to build two isolated coding exercises, but to study how different regression-based recursion rules behave under the same pricing environment.

The vanilla case is the main validation setting because it admits a direct one-dimensional American benchmark. In that setting, the Black–Scholes European put is used as a lower reference for the corresponding European claim, the implicit finite-difference solver is used as the main direct American benchmark, and the CRR tree is included as an auxiliary consistency check. This means the project does more than compare one regression method with another: it compares both Longstaff and TVR with a benchmark layer that solves the same vanilla American stopping problem under the same GBM assumptions.

The extension cases are handled differently. The Asian-style and jump-style studies are included as methodological extensions rather than as fully benchmarked replications of the same type as the vanilla put. Their role is to show how the two recursion methods behave once the state vector becomes richer or the simulator changes, not to carry the same direct-validation burden as the one-dimensional benchmark.

This distinction is important for interpreting the empirical evidence correctly: the vanilla study is the main validation layer, while the Asian and jump cases are structural tests of flexibility.

10.1 Convergence and Validation Perspective

A useful theoretical reference point from Longstaff–Schwartz is the lower-bound intuition behind the method. By Proposition 1 (LSM downward bias). *For any fixed finite basis and exercise design, the algorithm produces a specific stopping rule, so asymptotically its value cannot exceed the true American option value:*

$$V(X) \geq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \text{LSM}(\omega_i; M, K).$$

This statement is helpful conceptually, but it should be interpreted carefully in the present report. It is an asymptotic result about the true American value, not a guarantee that every finite-sample implementation must lie below a numerical finite-difference benchmark in every case. In the current results, Longstaff values can sit above or below the finite-difference reference depending on the case, which is consistent with the fact that the implementation uses a finite number of paths, a finite basis, and a numerical benchmark rather than the unknown true value itself. The main practical lesson is therefore not that the reported LSMC prices must always be conservative in finite samples, but that validation design matters if one wants the numerical output to reflect the intended lower-bound logic more faithfully.

A second Longstaff-Schwartz result is the practical convergence idea: under regularity conditions and with a sufficiently rich basis, the estimator can be made arbitrarily accurate as the number of simulation paths increases. In the paper, this is supported empirically by an out-of-sample diagnostic in which regression coefficients estimated on one path set are applied to a fresh set of paths. In our implementation, this idea directly motivates the two-pass validation design. The in-sample versus out-of-sample tables show that same-path valuation produces visible optimism for Longstaff, especially in the longer-maturity, higher-volatility cases, and that independent policy evaluation is therefore a substantive numerical correction rather than a cosmetic change.

The theoretical message from Tsitsiklis-Van Roy is different. Their framework emphasizes that approximation quality depends on the best projection error under the chosen basis and the distribution of representative states. A convenient way to summarize this is through the quantity

$$\epsilon_n^* = \min_r \|\Phi r - Q_n\|_{\pi_n} = \|\Pi_n Q_n - Q_n\|_{\pi_n},$$

which measures the best approximation error achievable at time step n under the state distribution π_n . The paper’s central theoretical point is that if representative states are sampled from the risk-neutral trajectory distributions, approximation errors remain controlled through the backward recursion rather than exploding with the horizon. In addition, as the number of simulated trajectories grows, the sample-based regression parameters converge to the parameters of the corresponding idealized population problem.

This perspective is directly relevant to the current implementation. It explains why the project uses shared simulated trajectory testbeds for both methods, why the TVR comparator is evaluated out of sample rather than reported only through fitted values, and why basis design and robustness checks are treated as substantive modeling choices. In the revised TVR results, the in-sample versus out-of-sample gap is much smaller than on the Longstaff side for the representative vanilla cases, which suggests that under the current shared design the fitted value-iteration scheme produces a comparatively stable stopping policy. This should not be read as a general theorem that TVR is always superior. It should be read as evidence that the two recursion schemes respond differently to the same validation framework.

The robustness studies are therefore part of the implementation argument, not an appendix of cosmetic sensitivity checks. On the Longstaff side, the study reports in-sample versus out-of-sample valuation, path sensitivity, exercise-grid sensitivity, basis sensitivity, and seed sensitivity. On the TVR side, the study reports in-sample versus out-of-sample valuation together with compact path, timestep, and basis sensitivity exercises. Taken together, these diagnostics show that path count alone does not determine pricing quality. Approximation space, exercise-grid design, and validation structure also influence the reported values and the implied stopping rule.

From the perspective of the report, the main implementation claim is therefore modest but meaningful. The project does not attempt to prove convergence numerically in the strict asymptotic sense of either paper. Instead, it uses the convergence and error-control ideas from the two papers to motivate a more careful empirical workflow: direct benchmarking where possible, out-of-sample policy evaluation, matched testbeds across methods, and explicit robustness analysis when approximation design may matter. That is the standard against which the implemented results should be interpreted.

11 Critical Assessment: Strengths, Weaknesses, and Practical Risks

The project highlights the strengths of both papers. Longstaff–Schwartz offers a direct and practical method for constructing an exercise rule from simulated cash flows. It is especially useful when the state process is rich enough that direct grids become inconvenient. Tsitsiklis–Van Roy provides a useful approximate-dynamic-programming perspective and emphasizes the recursive nature of value approximation. Together, the two methods illustrate two closely related ways of handling continuation values in American pricing.

The project also makes clear that both methods are numerically sensitive. Basis choice matters. Exercise-grid design matters. Validation design matters. The Longstaff results show that same-path pricing can be materially optimistic. The TVR results show that fitted value iteration can perform reasonably under the shared testbed, but its stability also depends on the chosen approximation space and on the distribution of simulated representative states.

These issues matter in practice. If regression-based American pricing methods are used carelessly, institutions can misprice early-exercise features through poor basis specification, unstable regression fits, weak benchmarking, or reliance on in-sample performance. The project does not claim to solve those governance problems. What it does show is that they arise naturally even in a controlled academic study.

Several simplifications remain. The finite-difference solver is a one-dimensional benchmark implementation rather than a production PDE framework. The Asian-style extension has no direct PDE benchmark. The jump process is stylized. Neither pricing framework is presented as a production library. These limitations should be kept in mind when interpreting the results.

12 Scope and Claims

The scope of the project is limited and should be stated plainly. This is a study of the Longstaff–Schwartz and Tsitsiklis–Van Roy papers with computational evidence under shared testbeds. Longstaff is implemented as the main practical pricing algorithm. TVR is implemented as a matched comparator under the same simulated cases and benchmark philosophy.

The project does not claim exact reproduction of every numerical example in either paper. The Asian-style and jump-style studies are shared extensions rather than exact replications of published tables. The project does not calibrate to market implied-volatility surfaces, does not incorporate a full production market-data workflow, and does not attempt to build a production pricing library.

Within that scope, the claims are clear. The project shows how the two papers can be understood through a common computational framework, how their recursion logic differs, and how those differences appear in benchmark alignment, out-of-sample behavior, and robustness.

13 Conclusions

This report studies American option pricing as an optimal stopping problem through two related regression-based frameworks. Longstaff–Schwartz serves as the main empirical pricing method, and Tsitsiklis–Van Roy serves as a matched approximate-dynamic-programming comparator under the same computational design.

The shared vanilla benchmark shows that both methods can produce reasonable American put values under a common GBM testbed, but that their benchmark behavior and validation sensitivity differ. Longstaff’s continuation regression on realized future cash flows leads to visible same-path optimism unless the policy is evaluated on a fresh sample. TVR’s fitted value iteration, when implemented under the same design and evaluated out of sample, produces values that are generally close to the finite-difference benchmark in the tested cases.

The shared Asian-style and jump-style studies show that both methods extend beyond the one-dimensional vanilla setting. The broader lesson is that regression-based American pricing is flexible, but its numerical performance depends on basis design, state representation, exercise-grid choice, and the way the stopping policy is validated.

14 Reproducibility / Code Execution

The Longstaff workflow is executed from the project root using the scripts under `src/LongstaffMethod`:

```
python src/LongstaffMethod/run_longstaff_vanilla.py
python src/LongstaffMethod/run_longstaff_asian.py
python src/LongstaffMethod/run_longstaff_jump.py
```

The TVR workflow is executed from the project root using the scripts under `src/TVR`:

```
python src/TVR/run_tvr_vanilla.py
python src/TVR/run_tvr_asian.py
python src/TVR/run_tvr_jump.py
```

Outputs are written to `outputs/`, visuals to `visuals/`, and revision notes to `SupportFiles/`. The vanilla bias note for Longstaff is stored in `SupportFiles/longstaff_vanilla_bias_fix_log.txt`.

The report can be rendered using Quarto:

```
quarto render report/american_options_lsmc_study.qmd --to pdf
```

If PDF rendering is unavailable, HTML output can be generated instead:

```
quarto render report/american_options_lsmc_study.qmd --to html
```

15 References

- Longstaff, F. A., & Schwartz, E. S. (2001). Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1), 113–147.
- Tsitsiklis, J. N., & Van Roy, B. (2001). Regression methods for pricing complex American-style options. *IEEE Transactions on Neural Networks*, 12(4), 694–703.